WORKED SOLUTION

STEP I 2009, Question 8

Pure • Euclidean geometry

STEP I 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q8

The key idea

A circle touches a line exactly when the distance from its centre to the line equals its radius. The circles C all have their centres on the line through the origin that bisects the angle between the x-axis and $3y = 4x$, so each of them touches both lines. That makes C the complete family of circles sitting in that corner, and the incircle in part (ii) must be one of them: we only need to pick the value of t that also makes the circle touch the third side.

Solution

Part (i)

The circle C has centre $P = (2t, t)$ and radius t. Since $t > 0$, the centre lies above the x-axis at a distance t from it, which equals the radius. So C touches $y = 0$, at the point $(2t, 0)$.

The line OP has gradient $\frac{t}{2t} = \frac{1}{2}$, so $\tan\alpha = \frac{1}{2}$ and

$$\tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3},$$

as required.

Now let β be the acute angle between the x-axis and the line $3y = 4x$, whose gradient is $\frac{4}{3}$. Then $\tan\beta = \frac{4}{3} = \tan 2\alpha$. Since $0 < \alpha < \frac{\pi}{4}$ (because $\tan\alpha = \frac{1}{2} < 1$), the angle 2α is also acute, and tan is one-to-one on acute angles, so $\beta = 2\alpha$.

? **Ask yourself** Why does $\beta = 2\alpha$ tell us that C touches $3y = 4x$?

The line OP makes angle α with the x-axis and angle $\beta - \alpha = \alpha$ with the line $3y = 4x$, so OP bisects the angle between the two lines. Dropping perpendiculars from P to each line gives two right-angled triangles with the same hypotenuse OP and the same

angle α at O, so the two perpendicular distances are equal, both being $OP\sin\alpha$. The distance to the x-axis is t, so the distance from P to $3y = 4x$ is also t, the radius. Hence C touches $3y = 4x$.

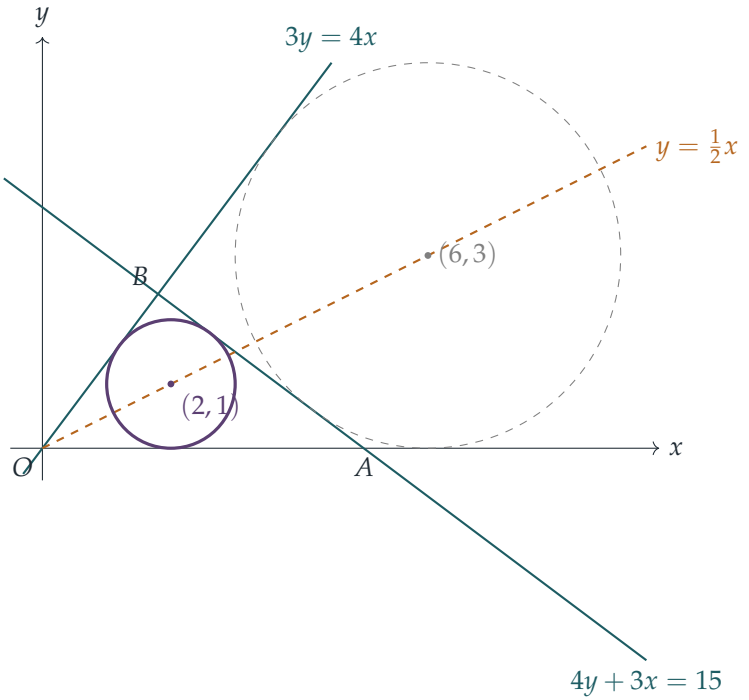
(As a cross-check, the perpendicular distance formula gives the distance from $(2t, t)$ to $4x - 3y = 0$ as $\frac{|8t-3t|}{5} = t$.)

Part (ii)

First we find the triangle. The lines $y = 0$ and $3y = 4x$ meet at $O = (0,0)$; $y = 0$ and $4y + 3x = 15$ meet at $A = (5,0)$. For the third vertex, substitute $y = \frac{4}{3}x$ into $4y + 3x = 15$:

$$\tfrac{16}{3}x + 3x = 15 \quad \Longrightarrow \quad x = \tfrac{9}{5}, \qquad y = \tfrac{12}{5},$$

so $B = (\frac{9}{5}, \frac{12}{5})$. The gradients of $3y = 4x$ and $4y + 3x = 15$ are $\frac{4}{3}$ and $-\frac{3}{4}$, whose product is -1, so the triangle has a right angle at B. Its sides are $OA = 5$, $OB = 3$ and $AB = 4$.



The incircle lies inside the triangle, so near O it sits in the acute angle between the positive x-axis and the line $3y = 4x$, touching both lines. Its centre is equidistant from the two lines, so it lies on the bisector of that angle, which by part (i) is the line $y = \frac{1}{2}x$ (the line OP). If the centre is $(2t, t)$ with $t > 0$, its distance from $y = 0$ is t, which must be the radius. So the incircle is one of the circles C from part (i).

It remains to make C touch the third side $3x + 4y - 15 = 0$. The distance from $(2t, t)$ to

this line must equal t:

$$\frac{|6t + 4t - 15|}{5} = t \quad \Longleftrightarrow \quad |10t - 15| = 5t.$$

Either $10t - 15 = 5t$, giving $t = 3$, or $15 - 10t = 5t$, giving $t = 1$.

Both circles touch all three lines, but only one lies inside the triangle. For $t = 3$ the centre $(6, 3)$ has $3x + 4y = 30 > 15$, so it is on the far side of AB from O (which has $3x + 4y = 0 < 15$): this is an *excircle*, the dashed circle in the figure. For $t = 1$ the centre $(2, 1)$ has $3x + 4y = 10 < 15$, and it is above $y = 0$ and below $3y = 4x$, so it is inside the triangle. A circle with its centre inside the triangle that touches all three sides lies inside the triangle, so the incircle has centre $(2, 1)$ and radius 1:

$$(x - 2)^2 + (y - 1)^2 = 1.$$

Checks

For a right-angled triangle with legs a, b and hypotenuse c, the inradius is $\frac{1}{2}(a + b - c)$; here $\frac{1}{2}(3 + 4 - 5) = 1$, as found. The inradius is also area/semi-perimeter $= 6/6 = 1$. Numerically, the incentre computed as the side-weighted average of the vertices, $\frac{4O+3A+5B}{12}$, comes out as $(2, 1)$ exactly, and the distances from $(2, 1)$ to all three lines are each 1.

Common slips

- Solving $|10t - 15| = 5t$ without the modulus and finding only $t = 3$, which is the excircle, not the incircle.
- Finding both roots but not explaining why $t = 1$ is the one inside the triangle.
- In part (i), saying $\tan\beta = \tan 2\alpha$ so $\beta = 2\alpha$ without noting that both angles are acute.

Beyond STEP

Every triangle has one incircle and three excircles, and the four centres are where the internal and external angle bisectors meet; the value $t = 3$ found here is the excircle opposite O. These circles lead into a rich vein of classical geometry, including the nine-point circle and Feuerbach's theorem that it touches all four. Coxeter's *Introduction to Geometry* is a classic place to explore this; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.